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HOW TO LEARN PROCESS-ORIENTED THINKING: AN EXPERIMENTAL INVESTIGATION OF THE EFFECTIVENESS OF DIFFERENT LEARNING MODES**

ABSTRACT

We examine the effectiveness of different learning modes in a process management context. We argue that many organizations will undergo a transformation from a function to a process orientation and that learning matters in this context, as a fundamental change in thinking is required. Using a questionnaire experiment with a sample of 147 participants, we found a significant influence of learning modes on the general success of learning process-oriented thinking. We observe that learning-by-doing is more effective than using documented knowledge. We raise awareness of the less studied underlying variables influencing the success of learning process-oriented thinking.

JEL Classification: D83, M10, M53.

Keywords: Business Process Management; Explicit Knowledge; Implicit Knowledge; Learning Modes; Process-Oriented Thinking; Questionnaire Experiment.

1 INTRODUCTION

Understanding how process-oriented thinking can be learned has major importance for research on strategy and organizations (Brazanga and Korac-Kakabadse (2000); Pathirage, Amaratunga, and Haigh (2007)). Process-oriented organizations are expected to be faster in delivering outputs, more adaptable to changes in the market, more responsive to the needs of customers, and superior in terms of quality (Hammer and Champy (1993); Braganza and Bytheway (1997)). Consequently, turning a company into a process-oriented organization should contribute to a company's success and therefore would represent a competitive advantage.

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However, although the available literature recognizes the advantages of a process orientation, most companies are still function-oriented (Brazanga and Korac-Kakabadse (2000); Vergidis, Turner, and Tiwari (2008)). This adherence to a function-orientation is primarily due to the fact that the knowledge required of process- and function-oriented organizations differs (Kugeler and Vieting (2011)). For example, in process-oriented organizations, while performing their work, employees without direct customer contact should be aware of the individual customer, but in a function-oriented organization the awareness of each individual customer is not required.

In general, in both function- and process-oriented organizations, although explicit knowledge might be present (Hawryszkiewicz (2010)), tacit knowledge (Howells (1996); Lam (2000)) has major importance, because it guides the daily work. Therefore, such knowledge must be acquired by the employees (Riege and Zulpo (2007)). Tacit knowledge is defined as implicit, individual-specific know-how and behavior (Leroy and Ramanantsoa (1997)) and it is "...typically acquired on the job or in the situation where it is used" (Sternberg (1994)). In contrast to explicit knowledge, which is codified and can be easily transferred (Nonaka (1991)), tacit knowledge relates to the individual and thus cannot be expressed and transferred that easily (Nonaka (1994); Nonaka, Takeuchi, and Umemoto (1996)). Hence, the acquisition of process-oriented tacit knowledge, which is necessary for the transformation from a function- to a process-oriented organization, requires that employees undergo fundamental change of thinking. Such a change of thinking is not easy to accomplish (Hammer and Champy (1993); Kilmann (1995)).

Despite the central importance of this research topic, previous research has not, to the best of our knowledge, looked at the learning modes that organizations could promote to enable such a fundamental change of mind. In the previous literature, Lam (2000) has discussed different learning modes that aim at enabling people to acquire tacit knowledge. These modes include personal exchange (Nonaka (1994); Snowden (1998)), learning-by-doing (Levitt and March (1988); Earl and Scott (1999)) and using explicit knowledge, for example, documented knowledge (Nonaka (1991)). Personal exchange can be promoted, for example, by actively setting up events and by allowing employees to share tacit knowledge in joint activities (Nonaka (1994)). Snowden (1998) proposes that tacit knowledge can be shared through psychosocial mechanisms and released through trust and its dynamics. Learning-by-doing instead requires the performance of a specific task that has to be learned, i.e., it takes place while acting (Paloniemi (2006)). As a consequence, it is the learners themselves who are responsible for the learning progress, and learning-by-doing often comes along with a high risk of failures as knowledge acquisition more or less takes place by a trial and error procedure (Levitt and March (1988)), thereby sometimes rendering efficient processes into inefficient ones. Using explicit knowledge requires that knowledge is externalized in publicly available sources like, for example, in documentations (Nonaka (1991); Snowden (1998)). Although we can find statements on the effectiveness of several learning modes in various studies, up until now we know of no empirical research that compares the effectiveness of different learning modes in a process management context.

It seems reasonable to assume that personal exchanges between employees take place in every organization, at least to some degree. Examples of occasions in which person-to-person exchanges take place are widespread, ranging from informal, unplanned sharing to exchanges in planned team meetings or in project teams. Clearly, an organization's transformation to a process orientation is accompanied by learning through personal exchange. In contrast, whether an organization focuses on supporting the acquisition of process-oriented knowledge through documented knowledge or by learning-by-doing is an active decision that has to be taken before the transformation begins. However, in the current literature this question remains open. This oversight is troubling, since anecdotal and empirical evidence both point to the fact that numerous organizations will undergo a transformation from function to process orientation (Lee and Dale (1998); Kumar, Movahedi, Lavassani, and Kumar (2010); Kohlbacher (2010); Kohlbacher and Gruenwald (2011)). Therefore, in this study we address this research gap by analyzing the effectiveness of the different learning modes in a process management context.

The paper is organized as follows. In Section 2, based on a review of the relevant literature, we derive our hypotheses. In Section 3 we describe our research methods and the results. Section 4 concludes with a discussion of the findings and a summary of the study's contributions to research and practice.

2 THEORETICAL FOUNDATIONS AND HYPOTHESES

Nordsieck (1931) and Kosiol (1934) introduced the basic idea of process orientation. Although researchers at that time concentrated on the functional structure of organizations, Nordsieck was the first to recognize processes as a basis for organizational structuring, i.e., he proposed to deal with operational and organizational structures separately. Kosiol (1934) concretized Nordsieck's idea by suggesting that positions in an organization should be defined by analyzing and clustering the single activities that are necessary to the process flow required by the particular work.

The definitions of process orientation now range from a narrow understanding that focuses on the management of business processes (Davenport and Short (1990); McCormack and Johnson (2001); Reijers (2006); Weske (2007)) to a broader understanding that centers on transformations in the organizational design (Osterloh and Hunziker (1998); Binner (2002); Fink (2003); Neumann, Probst, and Wernsmann (2003); Lockamy III and McCormack (2004); Osterloh and Frost (2006); Gaitanides (2007); Sens (2010)).

According to the first category of definitions, a process-oriented organization is one that is knowledgeable about end-to-end processes, i.e., it is aware that a process starts and ends with the customer. Hence, it aims at maximizing the value for the customer (Hammer (2010)). Such interpretations highlight that unlike function-oriented organizations, process-oriented organizations focus on process efficiency rather than on resource efficiency (Kugeler and Vieting (2011)). In particular, in such

narrow definitions, research regularly stresses the importance of determining process performance indicators, of measuring these indicators, and of controlling the resources that are affected by these measures (McCormack (2001)).

In contrast, according to the second category of definitions, process orientation refers to the general changes that accompany transformations in the organizational design and their effects on the organization (for example, on the strategic orientation or on management practices) (Chen, Tian, and Daugherty (2009)). Important examples of such transformation approaches are Business Reengineering (Hammer and Champy (1993)) and Business Engineering (Österle and Winter (2003)).

These broader definitions show that in process-oriented organizations, processes, rather than functional units, divisions, or departments, build the basis for management (Trkman, Stemberger, Jaklic, and Groznik (2007)). Further, these broader definitions show that process-oriented organizations are designed horizontally along the value-creating process rather than hierarchically (Kohlbacher (2010)), so that several business processes can be managed simultaneously (Harrington (1991)). Hence, authors who understand process orientation according to the broader category of definitions also acknowledge the points made by authors who have a narrower understanding. First and foremost, the interpretations of both categories have in common that they focus on process orientation as opposed to function orientation, and that they stress the importance of value creation for the customer.

In our study, we follow the narrow interpretation of process orientation and adopt the definition of Kohlbacher and Gruenwald (2011), which states that process orientation "...means focusing on business processes rather than emphasizing functional structure or hierarchy". For the management of business processes, this understanding means that managers should define the process model and should assign responsibilities based on this process model. Furthermore, this interpretation means that the aligned process goals should be formulated; these process goals serve as a basis for deducing concrete process measures within the operational management of these processes.

There is a broad consensus among researchers that whether a job is performed by someone who thinks in a function- or in a process-oriented way makes a difference (Nonaka (1991); Hammer and Champy (1993); Ambrosini and Bowman (2001)). Although the procedures of performing the work in a function- or in a process-oriented way are different, tacit knowledge, which unconsciously guides our daily work (Davenport and Prusak (1998); Stenmark (2001)) is important if employees are to be able to execute their tasks in both surroundings.

Given that a change of thinking is required for switching from a function to a process orientation, we assume that management's active support is conducive to employees' learning (Edwards, Braganza, and Lambert (2000)). However, although the advantages of a process-oriented organization are acknowledged in both science and practice, switching from a function- to a process-oriented structure involves serious challenges. In particular, such a switch can only be accomplished if employees learn how to think

in a process-oriented way (Kilmann (1995); Lucas (2010)). But because process-oriented knowledge is intangible, learning it is challenging. Thus, we assume that learning-by-doing might be a very effective method for learning how to think in a process-oriented way. In contrast, using documented knowledge seems to be less effective for promoting learning in this context, because a description of tacit knowledge can never be complete. It contains the necessary know-how only to a certain degree (Ungan (2006)). Given this background, we formulate the following hypothesis:

- H1.** *For learning process-oriented thinking to be successful, learning-by-doing is more effective than using process documentation.*

To better understand the success of learning process-oriented thinking, we must look at the major differences between function- and process-oriented organizations.

First, contrary to a function-oriented organization in which similar tasks are bundled in departments, a process-oriented organization establishes efficient value generating business processes (Kugeler and Vieting (2011)). (We note that we use the terms “process” and “business process” interchangeably.) A business process is defined as a set of connected activities that are necessary to deliver a designated business outcome (Davenport and Short (1990)). Within a business process, inputs are transformed into outputs by the resources of an organization. In such a process, the customers’ demands determine the desired business outcome, which in turn represents the starting point for the design of the business process. However, determining the sequence of activities within a business process is challenging and inevitably influences the success.

Second, within a function-oriented organization many departments are, often repeatedly, involved in the delivery of an order, which in turn results in many interfaces. These interfaces are inefficient, as additional effort is necessary, for example for the exchange of information, which causes waiting times to occur. But in a process-oriented organization, the bundling of similar tasks in functions is less important. Instead, in a process-oriented organization fewer employees work on one customer order (Kugeler and Vieting (2011)). Hammer and Champy (1993) point out that in good business processes, many formerly separate and distinct jobs and tasks are integrated and compressed into one. This statement points to the assumption that the assignment of roles to activities is also critical for success.

Last, another disadvantage of a function-oriented view is that employees who are involved in the production process often fulfill a customer demand without being responsible for the result. This lack of responsibility is due to the fact that functions regularly strive for defined goals, and in doing so ignore the stated goal of satisfying the customer. Ignoring this goal might even result in contradictory goals, such as high late-quarter sales compared to a continuous high workload of machines in production (Aoki (1986); Hammer and Champy (1993); Brazanga and Korac-Kakabadse (2000)). From a process-oriented point of view, activities’ specific goals are aligned with a superior process goal and employees are aware of this superior goal and of the overall context of their specific task (Kilmann (1995)). Here, we assume that the latter way of assigning goals is much more effective.

Given this background, we formulate the following hypotheses:

- H2.** *The learning mode makes a difference in terms of success of assigning activities.*
- H3.** *The learning mode makes a difference in terms of success of assigning roles.*
- H4.** *The learning mode makes a difference in terms of success of assigning goals.*

3 DATA AND METHOD

3.1 EXPERIMENTAL DESIGN

To address our research question, we conducted a questionnaire experiment in a German business school. In the course of this experiment, all participants had to perform a task that required process-oriented thinking. In this task, we required participants to serialize several predetermined activities, to assign roles to these activities, and to determine process goals. (See Appendix 1.) We chose a between-participants design. The treatment consisted of activating different learning modes before asking participants to perform the task.

In the first group (learning mode: “learning-by-documentation”) we presented participants with documented knowledge related to the task. (See Appendix 2.) We required participants in the second group (learning mode: “learning-by-doing”) to perform a sample task and gave feedback insofar as the sample solution was provided. (See Appendix 3.) We gave the third group of participants (control group: “acting from scratch”) no information at all. Participants were randomly assigned to one of the three experimental conditions.

3.2 MATERIAL AND MEASURES

Independent Measure: We included as an independent measure a variable consisting of three learning modes (acting from scratch, learning-by-doing, and learning-by-documentation).

We implemented *acting from scratch* by instructing the participants to perform the final task. We did so without providing them with any upfront information.

We implemented *learning-by-doing* by asking the participants to perform a task that was similar to the problem that they would have to solve later on. The two tasks differed only in the context of the business process. As a sample process, we chose the processing of loans, as we expected this context would be familiar for all participants. For each step in the task we provided participants with feedback by showing them the best-practice solution.

We implemented *learning-by-documentation* by providing participants with written information on the main ideas of function- and process-oriented thinking, and on the final process model for the loan process. As in the learning-by-doing condition, the final model of the loan process included a best-practice solution for the sequence of activities, the roles assigned to the activities, and for the process goals.

We note that we based the best-practice process on several in-depth discussions with experts in the process management field. We describe the best-practice process in Appendix 4.

Dependent Measures: The first dependent measure, *overall performance of process-oriented thinking*, related to the general success of participants in performing the task. We calculated this measure as the mean of three inferior performance measures that represent the other dependent variables, activities, roles, and goals. The process that the participants had to design in the course of the task was the process of handling exams in universities. We chose this process to ensure that the participants were at least somewhat familiar with the process.

In the course of performing the task, we first required participants to serialize 12 predetermined activities in a logical way. We measured the resulting *performance with regard to activities*, our second dependent variable, by comparing the results obtained to a best-practice process. We calculated deviations between the solution of the participants and the best-practice solution by using a conformance testing technique from the field of process mining. This technique makes it possible to measure whether the fitness of the observed process (in this case, the serialized activities of each participant) complies with the control flow specified by the process (the best-practice process model). We measure deviations from the best-practice process model and evaluate their impact. For example, a sequence of correctly ordered activities, defined according to the best-practice process, tends to be considered positively even when it is put in another position (Rozinat and van der Aalst (2005)). This procedure results in performance scores ranging from zero to one.

In our second step, we instructed participants to assign five predetermined roles to the 12 activities. To evaluate the performance of this sub-task, *performance with regard to roles*, our third dependent variable, we considered the number of reasonably assigned roles to activities compared to a predefined best practice, the total number of roles used, and the number of interfaces between roles. For the first criterion, we determined the number of correctly assigned roles. For roles being assigned exactly in the way as in the best-practice solution, participants were credited one point; for roles being assigned in the second-best way, participants were credited 0.5 points. We then divided the sum by 12 (as twelve roles had to be assigned to the respective activities in total). We computed the second criterion as one minus the percentage of the roles used compared to the maximum number available (i.e., compared to a maximum of five roles). We applied the same procedure for the last criterion, except that here, we divided the number of interfaces between the roles by 11, which is the maximum number of interfaces that can result from a process with 12 activities.

We determined *performance with regard to roles* by calculating the mean of the three criteria. Therefore, we defined this variable between zero and one.

In our third step, we asked participants to determine process goals. Thus, our fourth dependent variable relates to *performance with regard to goals*. We did not give participants the goals of the process, so the participants first had to determine the goals and then to assign them to each activity, while allowing for multi-activity goals and general process goals. To determine the performance score, we applied similar criteria as for the determination of the performance with regard to roles. We determined the percentage of activities that were linked to goals. In addition, we checked how many of the assigned goals were logical and finally, we checked if the goals assigned contradicted the generic process goals of “time, costs, and quality” (Lee and Dale (1998)). We then combined the three criteria into the measure *performance with regard to goals* by calculating the mean, which again results in a metric measure ranging from zero to one.

3.3 PARTICIPANTS AND PROCEDURE

In total, 165 students participated in our study. To determine whether the process-related description presented to participants in the learning-by-documentation condition was effective, we asked participants which is more important in a process-oriented organization, structuring an efficient value chain or the similarity of tasks. Of the 165 participants in our study, 51 had filled out the questionnaire in the learning-by-documentation condition. Out of these 51 participants, 49 (96.08%) correctly answered the question concerning importance. The others were eliminated. However, to ensure that the number of participants was equally distributed among the three conditions reflected in the questionnaires we randomly chose another 16 questionnaires to exclude from the sample. Doing so resulted in our final sample of 147 undergraduate management students.

The sample consists of 103 men (70.1%), 40 women (27.2%), and four participants (2.7%) who did not indicate their sex in the questionnaire. Participants' ages range from 20 to 32 years; the mean age is 23.04 years ($SD = 2.25$; $N = 140$). One hundred fourteen (77.6%) of the participants had professional experience, 29 (19.7%) had no professional experience, and four (2.7%) did not answer the question, Professional experience ranges from zero to 144 months ($M = 26.8$ months; $SD = 31.85$; $N = 143$).

We distributed questionnaires to students at a German business school that specializes in finance and management. We asked the participants to fill out the questionnaire anonymously and without consulting others. We instructed participants to thoroughly consider the information given, to visualize the situation as if it were real, and to perform the task. Finally, we asked them to provide their demographic details. On completion, we debriefed and thanked them. All the students participated voluntarily without being paid.

3.4 DATA ANALYSIS

We test our hypotheses by performing a one-way analysis of variance (ANOVA) with the learning mode as independent variable. We test the influence for the overall performance as well as for the performance relating to activities, roles, and goals. To test for statistically significant differences between the learning modes we use the Games-Howell post-hoc test, which does not rely on homogeneity of variances, and the Tukey's Honestly Significant Difference (HSD) post-hoc test, which relies on homogeneity of variances.

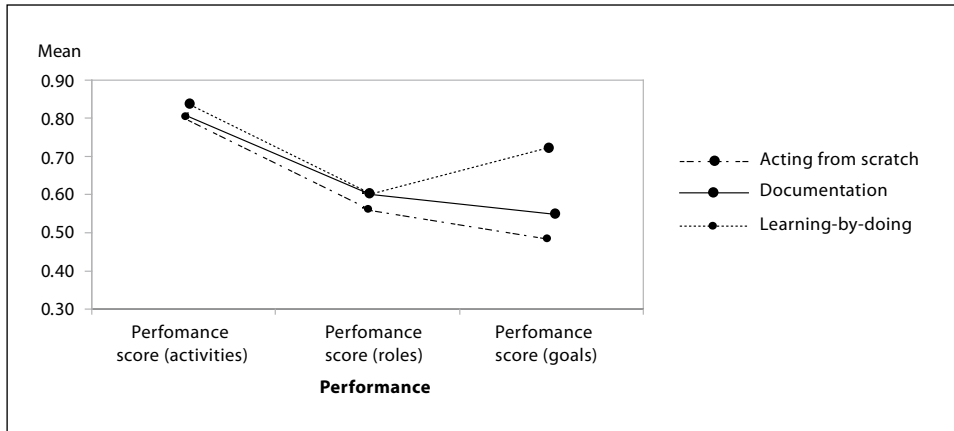
4 RESULTS

Table 1 presents the mean values and standard deviations of the performance scores as determined over the samples of the different learning modes. The table shows that the overall performance of process-oriented thinking and all but one of the concrete performance measures are lowest in the acting-from-scratch condition. The table also shows that the overall performance of process-oriented thinking is highest in the learning-by-doing condition ($M = 0.73$, $SD = 0.10$) compared to the acting-from-scratch ($M = 0.60$, $SD = 0.12$) and documentation ($M = 0.66$, $SD = 0.13$) conditions. This observation is also valid for all but one of the concrete performance measures: only for performance related to roles do we see that documentation ($M = 0.61$, $SD = 0.11$) is slightly better than learning-by-doing ($M = 0.60$, $SD = 0.10$). Thus, on a descriptive level, the table mostly confirms the hypothesized pattern.

Table 1: Means and Standard Deviations for Performance in the Three Learning Mode Conditions

	Learning mode					
	Acting from scratch		Documentation		Learning-by-doing	
Performance	Mean	SD	Mean	SD	Mean	SD
Overall performance	0.60	0.12	0.66	0.13	0.73	0.10
Performance with regard to activities	0.81	0.08	0.81	0.07	0.85	0.06
Performance with regard to roles	0.56	0.11	0.61	0.11	0.60	0.10
Performance with regard to goals	0.44	0.34	0.55	0.35	0.73	0.29

Figure 1 shows the relation between learning modes and performance criteria, thereby illustrating the descriptive results contained in *Table 1*.

Figure 1: Performance in the Three Learning Mode Conditions

We conducted a one-way ANOVA using learning mode as the independent variable. The results of the ANOVA show a statistically significant influence of learning modes on the success of learning process-oriented thinking, $F(2, 144) = 13.321, p < 0.001$. We also observe a significant influence of the learning mode on the performance of serializing predetermined activities: $F(2, 144) = 4.566, p < 0.02$, of assigning roles, $F(2, 144) = 3.161, p < 0.05$, and of assigning goals, $F(2, 144) = 9.536, p < 0.001$.

Hypothesis H1, which states that learning-by-doing is superior to using documentations, is confirmed (Games-Howell post-hoc test, $p < 0.001$). We observe a statistically significant difference between the learning mode related to the success of serializing predetermined activities (Games-Howell post-hoc test, $p < 0.02$), indicating that participants in the learning-by-doing condition perform better than do those in the learning-by-documentation condition. Thus, we can confirm H2. Also, we see that participants in the learning-by-doing condition perform better than do those in the learning-by-documentation condition affecting the assignment of goals (Games-Howell post-hoc test, $p < 0.03$), thus providing support for H3. Only H4 cannot be confirmed, as learning-by-doing is not significantly better for the assignment of roles (Tukey's HSD post-hoc test, *ns*).

Finally, we compared the performance of participants in both learning mode conditions with the performance of participants in the acting-from-scratch condition (i.e., the control group). Participants under the learning-by-doing condition performed significantly better (Games-Howell post-hoc test, $p < 0.001$) than did those who were acting from scratch. For participants under the learning-by-documentation condition the results are not statistically significant (Games-Howell post-hoc test, *ns*).

5 DISCUSSION AND CONCLUSION

Our research investigates the effectiveness of different learning modes in a process management context. We show that for learning process-oriented thinking to succeed, learning-by-doing is more effective than is using documented process knowledge. Our results indicate that learning-by-doing is more effective for the success of serializing predetermined activities and for assigning goals, but less effective for the assignment of roles. These findings might be indicative of the varying complexity of the three tasks that are related to the subperformance measures. The less structural information we provided, the worse our participants performed. This effect leads us to assume that the low level of information provided results in an increasing abstractness and difficulty of the tasks to be performed. In our experiment, the task of assigning activities was well structured, since we provided predetermined activities that had only to be serialized. Hence, no effort was needed to decide which activities to use, because we instructed participants to use all of them. But for the second task, the assignment of roles, we provided our participants with several roles and instructed them to decide whether to use all of these roles or just a few of them, depending on what they thought was more logical for performing the task successfully. For the third task, the assignment of goals, we gave no information at all, so that the task was totally unstructured. We find that the more information is given to participants, the better they perform in general. The only exception appears in the learning-by-doing condition. Here, participants perform better in the third task than in the second, and the superior results, as compared to the other learning modes, are especially high for this performance measure. This result is particularly interesting, because the third task was the least structured, thus requiring the deepest understanding of our participants. Here, we can assume that people who learn process-oriented thinking by doing are better in transferring knowledge to unfamiliar situations than people who learn process-oriented thinking by using documented process knowledge.

The generally superior results of our participants in the learning-by-doing condition might be due to the high involvement in performing the task. Participants in the learning-by-doing condition were more or less forced to perform the task consciously, but participants in the learning-by-documentation condition could either not take the task so seriously or could have taken a more distant attitude to the documented knowledge. In sum, using documented knowledge seems to be less helpful in learning how to do something and how to proceed in an uncertain environment. Thus, it seems to be less suitable for learning process-oriented thinking. As a consequence, learning-by-doing rather than learning-by-documentation should be actively promoted by organizations that are changing from a function to a process orientation.

There are several limitations to this study's contributions. One is that the questionnaire experiment is accompanied by the potential problem that participants perform a hypothetical task rather than behaving in a real-life situation. However, a questionnaire experiment also has some advantages. First of all, it allows the researcher to explicitly control certain variables that could have a causal effect on the dependent variable, i.e., on the success of learning process-oriented thinking. In general, we expect this

possibility to control specific variables to result in a more efficient design and thus to less ambiguous findings (den Dulk and de Ruijter (2008)). Closely related to this point is that the systematic treatment endemic in experiments enables the analysis of causal relationships. Given this background and taking into consideration that the effectiveness of learning modes has not been analyzed so far, we believe that our research method is appropriate for analyzing our research question.

An additional point that merits attention is that we approached our participants in the course of a lecture. Although the participation in our study was voluntarily, we cannot rule out that some participants only agreed to participate in our study because of peer pressure, or because they thought that their professors expected them to do so.

There are also limitations in the survey itself. First, because we used a between-participants design, although in reality employees might not only be exposed to one specific learning mode but instead learn, for example, by doing and by using documented knowledge at the same time. Our reasons for choosing a between-participants design were twofold. On the one hand, we wished to avoid overloading the questionnaire with too much information and with too many questions. On the other hand, we were concerned that including different learning modes would make it impossible to isolate the effects stemming from the single learning modes. A second limitation is that the given task only encompassed one particular business process. A further limitation is that for the evaluation of the order of activities within the task we used only one best-practice process. However, we cannot rule out that there are several correct possibilities for performing this subtask. Nevertheless, the quality of the best-practice process was evaluated by employees of the business school who were familiar with the processing of exams as well as by experts in Six Sigma (a common process-oriented improvement method). Additionally, the conformance technique allows for a precise calculation of different deviations from the best-practice process.

For the scientific community, the results of our investigation are important, as empirical research on organizations' transformations to a process orientation is still rare (for a notable exception see Mc Nulty and Ferlie (2004)). Benner and Tushman (2003) highlight that "...much previous literature is prescriptive and aimed at educating managers in implementing process management practices" and that "...process management's proponents have promoted process improvement practices as universally beneficial for organizations". In line with this reasoning, the previous literature acknowledges that the understanding required of process-oriented and function-oriented organizations differs (Kugeler and Vieting (2011)), but the effectiveness of different learning modes has not been investigated in a process management context before. Indeed, although there is an extensive body of research on learning in general and on different learning modes, our study is one of the very few to empirically analyze the learning modes in this specific research context. Given the fact that many organizations are expected to undergo a transformation from a function-oriented to a process-oriented structure within the next few years, such an investigation seems long overdue. This is not to say that process orientation is a universal recipe. Although it is very often recommended in the current literature that organizations should strive

for a strict process orientation, a combination of functional and process orientation might be the best design strategy for some organizations (Picot and Franck (1995); Brazanga and Korac-Kakabadse (2000)). Further, some earlier problems, for example, the indivisibility of critical resources in a multiproduct firm, might even require a firm to stick to a mere function orientation. However, due to the reasons mentioned above, transforming an organization's structure from function-oriented to process-oriented seems to be very beneficial in many cases.

Our findings also have important implications for academic teaching of organizational design. First, our findings should be included in academic teaching by reporting them directly to students who might be involved in, or even be responsible for, the successful implementation of organizational transformations later on in their careers. Students might profit from interdisciplinary lectures in which findings on individual and organizational learning and business process management research are discussed. Second, since our results indicate that learning-by-doing is the most promising learning mode for learning how to think in a process-oriented way, academic teaching of organizational design should focus on this learning mode for conveying the relevant contents of this research domain. This suggestion might be implemented by, for example, using case studies more intensively.

For practitioners, the results of our study are particularly relevant for the learning modes that should be promoted. Our results raise awareness of the so far less well studied underlying variables that influence the success of transforming a process-oriented organization, and which allow for drawing conclusions on the promotion of process-oriented thinking. Our study stresses that learning matters, and that organizations should actively promote primarily learning-by-doing and to a lesser extent learning-by-documentation.

Although our study offers a range of practical and theoretical implications, there are still some open questions remaining. On the one hand, to enlarge the generalizability of our results the questionnaire experiment might be performed for other business processes. On the other hand, the focus on the real world should be enhanced. Although we tried to integrate such a focus by choosing real-world business processes and by asking participants to visualize the situation as if it were real, the real-world focus could be enhanced, for example, by conducting field experiments.

APPENDIX

APPENDIX 1: VIGNETTE USED IN ALL EXPERIMENTAL CONDITIONS (I.E., FOR ACTING FROM SCRATCH, LEARNING-BY-DOCUMENTATION AND LEARNING-BY-DOING)

Imagine that you work at a university. As part of your job, you have to design the examination process of your university as efficiently as possible.

To fulfil this task, please perform the following steps one after the other:

- 1) First of all, please order the activities which are listed on the next page in a logical way. Please use *all* activities and enter the respective numbers in the boxes on the next page.
- 2) In a second step, the process needs to be structured. To structure the process, please assign roles to the activities. Letters have been assigned to the roles which are listed on the next page. Please write the respective letters on the lines below the boxes for the activities. You are allowed to assign the same role to several activities. There is no need to use all roles that are proposed in the list; please do only use those roles that seem appropriate to you.
- 3) In the third step, please determine goals for the activities. The goals should be challenging so that there is an incentive for performing the activities as efficiently as possible. You can assign the same goal to several activities.

Please perform the tasks mentioned above as well as possible.

The following roles and activities are available:

Activities:

- | | |
|------------------------------|---------------------------------|
| To sort exams (1) | To copy exams (7) |
| To control grading (2) | To mark exams (8) |
| To design exam (3) | To publish grading (9) |
| To record grading (4) | To adjust layout (10) |
| To design exam exercises (5) | To prepare sample solution (11) |
| To hand out exams (6) | To collect exams (12) |

Roles:

- | | |
|------------------------|-----------------------|
| Lecturer (A) | Secretariat (D) |
| Examination office (B) | Student assistant (E) |
| Invigilator (C) | Students (F) |

Goals	—	—	—	—	—	—	—	—	—	—	—	—	
Activities		→		→		→		→		→		→	
Roles	—	—	—	—	—	—	—	—	—	—	—	—	

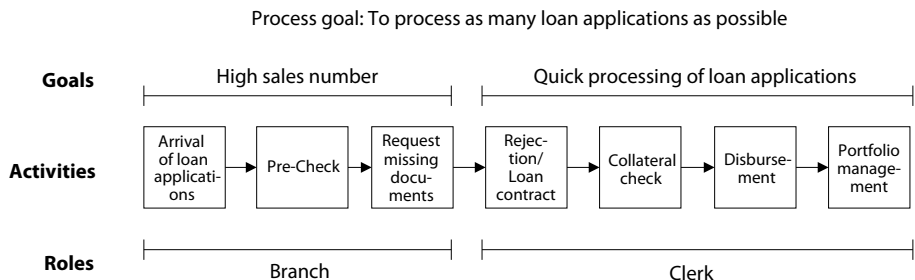
APPENDIX 2: ADDITIONAL INFORMATION IN THE LEARNING-BY-DOCUMENTATION CONDITION

In the following, you are asked to design the examination process of a university as efficiently as possible. To be able to fulfil this task as requested, it is inevitable that you familiarise yourself with the fundamental idea of process management.

Traditionally, companies are organized primarily in a *function-oriented* way, i.e., similar tasks are bundled in departments. For example, in the marketing department, all activities, which might be necessary for the sale and delivery of different products and services, might be combined. Within a function-oriented organization, a lot of departments are (often repeatedly) involved for the delivering of an order, which in turn results in many interfaces: For example, on the way from an order to the delivery of a final product, marketing, engineering, production, purchasing department, etc. (often repeatedly) have to deal with this order one after another. These interfaces are very inefficient because overcoming them might lead to additional expenses (for example, for informing the employees involved and for the exchange of information), waiting times and errors. This limitation can be addressed by a process-oriented organization, in which efficient value generating business processes are established and less employees work on one customer order.

In a *process-oriented* organizational structure, organizational units are formed in a way that allows for establishing value generating processes. That is, instead of bundling similar tasks, one focuses on the value chain. In particular, a business process consists of a set of connected activities, which are necessary to deliver a defined business outcome (e.g., the processed loan application). Each business process has a superior process goal (such as, for example, satisfying the customer), which is binding for all persons being involved in the different activities of the business process. Specific goals of activities should be aligned with this superior process goal.

In the following figure, the idea of process orientation is visualized based on the example of a loan application process in banks.



Please answer the following questions to control yourself by marking the correct answer:

Which of the following statements is right?

- ☐ In a process-oriented organizational structure, the focus is on an efficient value chain.
- ☐ In a process-oriented organizational structure, the focus is on the similarity of tasks.

Now that you are familiar with the idea of process management, please move on to the next side and perform the task given there as well as possible.

APPENDIX 3: ADDITIONAL INFORMATION IN THE LEARNING-BY-DOING CONDITION

In the following, you are asked to design the examination process of a university as efficiently as possible. To be able to fulfil this task as requested, we would like to give you the opportunity to gather some experience in structuring processes, in particular based on the example of a loan application process.

Please order the following activities in a logical way. Please use all activities and enter the respective numbers in the boxes.

Activities

Portfolio management (1)

Request missing documents (2)

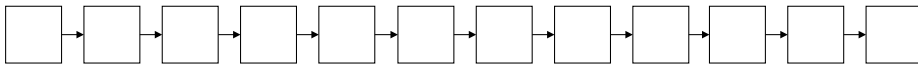
Disbursement (3)

Rejection/Loan contract (4)

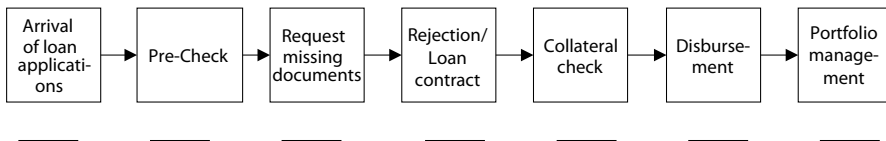
Pre-Check (5)

Arrival of loan applications (6)

Collateral check (7)



To ensure a maximal learning effect, we would like to show you a best-practice solution. Please have a look at this solution, without making any changes to your own solution. This exercise is for practice only, so that there is no need to worry whether you solved the task correctly.



For structuring the process, please assign roles to the activities. Letters have been assigned to the roles which are listed below. Please write the respective letters on the lines below the boxes for the activities. You are allowed to assign the same role to several activities. There is no need to use all roles that are proposed in the list; please do only use those roles that seem appropriate to you.

Roles:

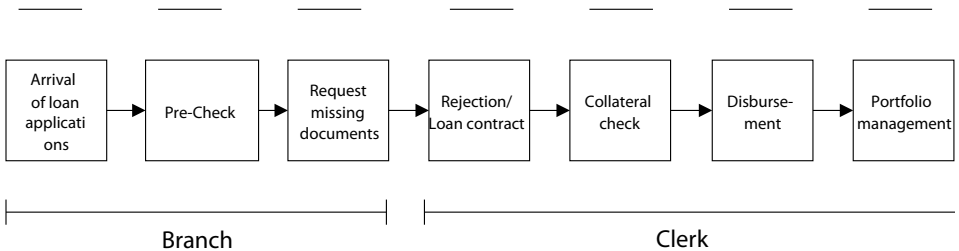
Branch (A)

Clerk (C)

Clerk Manager (B)

Portfolio management clerk (D)

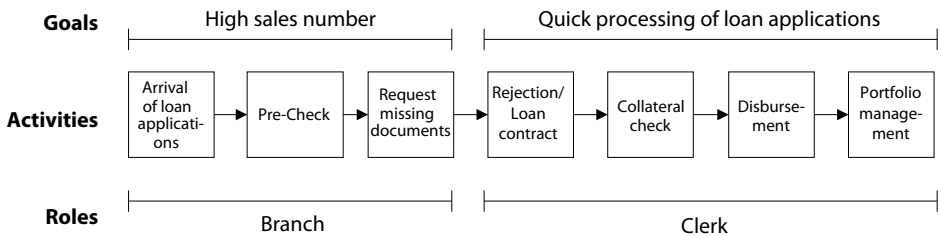
Again, we would like to show you a best-practice solution. Please have a look at this solution, without making any changes to your own solution.



In the third step, please determine goals for the activities. The goals should be challenging so that there is an incentive for performing the activities as efficiently as possible. Please note the determined goals on the lines above the activities. Examples for goals are “to perform the task as fast as possible” or “to maximize the employee efficiency”. You can assign the same goal to several activities.

Again we would like to show you a best-practice solution. Please have a look at this solution, without making any changes to your own solution.

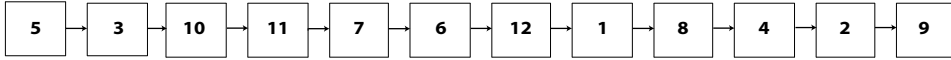
Process goal: To process as many loan applications as possible



Now that you have gained some experiences in structuring processes, please move on to the next side and perform the task given there as well as possible.

APPENDIX 4: BEST-PRACTICE SOLUTIONS FOR EVALUATING TASK PERFORMANCE

Best-practice process as a basis for evaluating performance with regard to activities:



Best-practice solution as a basis for evaluating performance with regard to roles:

Number of activity	Roles to be assigned to the activities
1 (To sort exams)	B (also logical: A, D, E)
2 (To control grading)	A
3 (To design exam)	A
4 (To record grading)	A (also logical: B, D, E)
5 (To design exam exercises)	A
6 (To handout exams)	B (also logical: C)
7 (To copy exams)	B (also logical: D, E)
8 (To mark exams)	A (also logical: E)
9 (To publish grading)	A (also logical: B, D)
10 (To adjust layout)	A (also logical: B, D, E)
11 (To prepare sample solution)	A
12 (To collect exams)	B (also logical: C)

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